

EXPERIMENTAL STRESS ANALYSIS

PROFESSIONAL ELECTIVE-I

VI Semester										
Course Code		Category		Hours / Week			Credits	Maximum Marks		
A6AE28		PEC		L	T	P	C	CIE	SEE	Total
				3	0	0	3	40	60	100
UNIT-I		MEASUREMENTS AND EXTENSOMETERS								
Principles of measurement, Accuracy, Sensitivity, Range, Types of Error, Mechanical, Optical, Acoustical and Electrical extensometers and their use, advantage and disadvantage.										
UNIT-II		STRAIN GAUGE – PRINCIPLES AND STRAIN MEASUREMENT								
Principles and operation of electrical strain gauge, Requirement, Type and their uses, Material for strain gauge, Calibration, Bridge sensitivity, Wheatstone bridge and potentiometer circuits for static and dynamic strain measurements, Strain indicator, Rosette Analysis.										
UNIT-III		PHOTOELASTICITY, FRINGE INTERPOLATION TECHNIQUES								
Concept of Light, Two dimensional Photo elasticity, Photo elastic effects, Stress optic law, Plane and circular polariscope, Interpretation of fringe pattern, Compensation and separation techniques, Photo elastic material.										
UNIT-IV		NON-DESTRUCTIVE TESTING-I								
Fundamentals of Non Destructive Testing, Radiography, Ultrasonic Inspection, Ultrasonic C-Scan, Magnetic particles Inspection, Fluorescent penetrant technique, Eddy current testing, Acoustic Emission Technique.										
UNIT-V		NON-DESTRUCTIVE TESTING -II								
Fundamentals of brittle coating methods, Analysis of brittle coatings, Industrial application of brittle coating, Introduction to Moiré Techniques, Holography, Thermography.										
Text Books:										
1. Daily J. W, Riley W. F (2005), Experimental Stress Analysis, 4th edition, McGraw- Hill, New Delhi. 2. Thomas G. Beckwith, Maragoni, Lienhard (2009), Mechanical Measurements, 6th edition, Pearson Education, New Delhi.										
Reference Books:										
1. Sadhu Singh (2009), Experimental stress Analysis, 3rd edition, Khanna Publications, New Delhi. 2. Prasad (2011), Non- Destructive Test and Evaluation of Materials, 1st edition, Tata McGraw-Hill, New Delhi.										
COURSE OUTCOMES : STUDENTS SHOULD BE ABLE TO:										
1. Understand the basic concepts of experimental stress analysis 2. Demonstrate the principles of major types of extensometers 3. Apply the knowledge of Strain gauges in aeronautical domain 4. Understand the principles of Rosette analysis and fringe techniques 5. Understand NDT techniques used in the structural analysis										